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Procrastination Root Cause Identifier: A Dual-Model AI-Driven Conversational Framework for Multi-Factor Behavioural and Psychological Analysis

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Abstract

Procrastination is a widespread pattern that often impacts enthusiasm, psychological well-being, and efficiency. Understanding the root causes of procrastination is essential because it typically stems from a combination of psychological, emotional, and environmental factors. In this work, Procrastination Root Cause Identifier (PRCI) has been introduced, an AI-based chat system that uses conversations to explore why someone procrastinates, including reasons not linked to disorders. Natural language processing (NLP) and machine learning (ML) techniques are used by the system to analyse user inputs to find potential underlying causes, such as dopamine-driven behaviour, perfectionism, fear of failure, lack of interest, environmental distractions, and feelings of overwhelm. PRCI offers a comprehensive language model API connected to an interactive and empathetic chat experience developed with Streamlit to ensure responsive and natural conversation. A visual dashboard that shows the analysis results, including probabilities for problem detection and likelihoods of behavioural causes, can help users effectively assess their procrastination patterns. The backend uses a traditional machine learning pipeline that uses TF-IDF and Logistic Regression for multi-label classification of root causes, in addition to a refined BERT model with dual outputs for screening for anxiety and depression. To improve accuracy and interpretability, the system was refined through extensive parameter tuning after being trained on synthetic datasets. Experimental results show that PRCI can effectively differentiate between disorder-related and behavioural factors contributing to procrastination, providing users with personalized insights for self-reflection and improvement.
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Keywords: Procrastination; behavioural root cause; multi-factor analysis; dual-model reasoning; depression screening; anxiety detection; BERT model; TF-IDF; mental health support; deep learning; user interaction analysis

1. Introduction

Procrastination is a common behaviour where people delay tasks they intend to complete, even when they know it could lead to negative outcomes. It impacts various aspects of life, including school, work, and personal matters. It often causes situations to worsen, makes stress feel heavier, and can even contribute to mental health concerns. Research has linked procrastination to several psychological and emotional factors, such as anxiety, perfectionism, fear of failure, and lack of motivation. Because these influences interact in complex ways, understanding procrastination requires multi-factor analysis, rather than examining a single dimension in isolation. However, identifying the root causes of procrastination is complex, as it typically arises from a blend of internal dispositions and situational triggers. Earlier studies often examined either disorder-related or behavioural reasons in isolation, which leaves a gap in understanding how both interact in real scenarios. This motivates the need for an approach that can observe these factors together rather than separately. To address this gap, Procrastination Root Cause Identifier (PRCI) has been developed, to perform dual-model reasoning by integrating deep learning with traditional machine learning techniques. The main contribution lies in presenting a dual-analysis framework that simultaneously screens potential disorders and behavioural triggers, providing a more complete view than traditional single-focus methods. PRCI employs deep learning and traditional machine learning techniques to interpret user interactions and derive insights into behaviour.

The system consists of two complementary modules: a fine-tuned Bidirectional Encoder Representations from Transformers (BERT) model and a traditional Machine Learning (ML) pipeline using TF–IDF vectorization and Logistic Regression. The BERT model includes dual output layers for screening depression and anxiety, enabling the system to highlight early mental-health-related signals based on language cues. The ML pipeline identifies six non-disorder root causes—perfectionism, fear of failure, lack of interest, environmental distractions, dopamine-driven habits, and emotional states—reflecting common real-life procrastination patterns. The combined approach overcomes limitations seen in earlier sentiment or emotion-based model by offering both contextual depth and interpretability. Synthetic yet meaningful datasets were developed for both components. Depression and anxiety samples were labelled according to commonly observed linguistic patterns, while the behavioural datasets captured realistic procrastination scenarios across diverse emotional and situational contexts. Text pre-processing included tokenization, normalization, stop word removal, and lemmatization to ensure clean and consistent inputs. The BERT model was fine-tuned with adaptive learning rate scheduling, dropout regularization, and the AdamW optimizer to reduce overfitting, while hyper-parameter for the Logistic Regression model were tuned using grid search to improve multi-label classification performance. PRCI was evaluated using established metrics such as Precision, Recall, F1-score, and Accuracy, along with True Positive (TP), False Positive (FP), and False Negative (FN) counts for clarity. ROC–AUC curves were used to assess model stability across thresholds [1], [2]. The evaluations ensure that both deep learning and traditional ML components perform reliably and meet expectations for systems supporting behavioural and mental-health research.

The user interface, built with Streamlit, features a conversational chatbot powered by an LLM API, offering supportive and natural responses. A visual dashboard presents disorder probabilities and behavioural cause likelihoods through progress bars, radar charts, and probability gauges, allowing users to easily understand their results. All interactions are processed locally to maintain privacy and uphold ethical standards. PRCI's dual-analysis design distinguishes it from earlier tools that rely solely on surveys or single-category disorder detection. Its ability to connect predictions with specific linguistic cues also addresses the common "black-box" concern in psychological AI systems. In addition, the modular structure supports future extensions such as multimodal inputs (speech, facial expressions, typing patterns), making it suitable for broader wellness applications across educational, workplace, or mental-health settings.

Ethical considerations remain central to PRCI's development. The tool functions as a self-reflective support system rather than a diagnostic instrument. It highlights possible habits and emotional patterns that may contribute to procrastination while avoiding clinical claims. With a focus on clarity, transparency, and privacy, PRCI aligns with recommended practices for AI systems operating in sensitive domains. The rest of the paper is organized as follows: the methodology and model design are presented in Section II, experimental evaluation is discussed in Section III, and Section IV concludes with insights and future directions.

2. Literature Review

Research on procrastination has changed over time. It used to focus on ideas about why people procrastinate. Recently, studies now include technology tools to help reduce procrastination. In the past, work like Costa's Procrastinate No More used simple models and personal questions to find patterns [3]. It tried to give personalized advice based on user habits. However, this method relied on users to report their habits. It did not use proven therapy methods. This limited its ability to predict problems or suggest good solutions. Our study builds on this work. We include behavioural data, sentiment analysis of journal entries, and strategies based on cognitive behavioural therapy. These steps create a system that is more flexible and based on psychological methods. Ibrahim et al. [4] demonstrated of studies about AI tools that help with procrastination. The finding show that most tools focus on chatbots, productivity tools, and helping with schoolwork. These tools help with managing time but do not include psychological or therapy methods. In a controlled experiment, Lee et al. [5] created Moa, a chatbot-based to-do app that help users to set goals and manage their actions. The study used a randomized controlled trial that improved user time management and reduced stress. However, the system relied on user reflection, which limited its long-term use. The new system uses two models. One predicts user behaviour, and the other provides conversation-based support. This setup aims to keep users engaged longer and offers more personalized help. Similarly, Imhof et al. [6] tested different machine learning models, including Random Forest, Gradient Boosting, SVM, and Neural Networks. They wanted to predict procrastination during online learning. The results showed that ensemble models, like Random Forest and Gradient Boosting, worked best. But their models did not include methods for helping users change behaviour. Our study combines these prediction models with cognitive behavioural therapy (CBT) conversations. This helps turn detection into action, providing both insight and intervention. Ghasempour et al.[7] found that procrastination is linked to feelings of anxiety, depression, and low self-esteem. The study used statistical analysis but did not suggest ways to fix these issues. Other studies that look at stress and procrastination at a single point in time [8] also find links between these problems. These studies do not explore technology to help. Provoost et al. [9] reviewed chatbots used for changing behaviour. They showed that these tools are easy to access but not very personalized and work best in the short term.

Fitzpatrick et al. [10] created Woebot, which uses the principles of cognitive-behavioural therapy (CBT). They showed that CBT techniques can be delivered through chat to help reduce depression and anxiety. But both tools did not have special features for procrastination or for long-term behaviour change. Further, YOLO Procrastination Preventor [11], using YOLOv5 and OpenCV, detects when someone uses a smartphone in real time. Then, it sends an alert to stop the distraction. This method helps control external cues but does not target internal reasons like perfectionism or fear of failure. Steel et al. [12] found that impulsivity, self-control, and habit development are important parts of procrastination. These studies used tests and data analysis to support their findings. They provide a base for new AI tools that predict behaviour and help with therapy. They can change based on how a person acts over time. Tuckman et al. [13] created a system that uses both BERT and Bi-LSTM to find signs of depression and anxiety in social media texts. The system was accurate in tests. However, the study was later withdrawn because others could not repeat the results. This shows that AI tools need to be clear and based on real clinical understanding. Lally et al. [14] showed that models like BERT and RoBERTa perform well at identifying mental health issues from text. But these models mainly only classify data. Further, Therapeutic chatbot design has been studied using rule-based systems. One chatbot using AIML combined structured cognitive-behavioural therapy (CBT) responses with basic emotion detection [15], [16]. This system did not have the ability to change its responses based on user input or learn from data. Insel et al. [17] introduced digital phenotyping that tracks behavioural and emotional data from smartphones and sensors

continuously. The framework supports passive mental health monitoring but does not offer real-time help. Moreover, study has been conducted on identifying health issues from clinical notes [16] that uses traditional machine learning models along with deep learning. This improved the reliability of the classification. The approach uses both methods, one part uses simple techniques that are easy to understand and the other uses advanced models to understand context. Unlike earlier applications that only classified data, this system also offers suggestions for behavioural change. It uses understandable AI that can be shown to users.

3. Methodology

The Procrastination Coach system was designed as an intelligent companion blending conversational empathy with machine learning–driven behavioural insight. The architecture integrates three core components: a Streamlit-based conversational interface, a BERT-based model for disorder screening, and a classical machine-learning classifier for identifying root causes of procrastination such as perfectionism or low motivation as shown in Fig.1. Together, these modules process user dialogues, analyse underlying emotional cues, and provide psychologically informed feedback in real time.

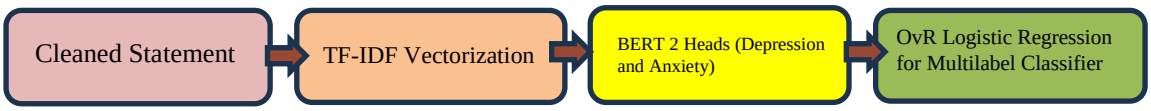

Fig.1. End-to-End processing pipeline

In this work, a behavioural root cause refers to a non-clinical pattern (such as perfectionism, fear of failure, lack of interest, environmental distraction, dopamine-driven habits, or feeling overwhelmed) that contributes to procrastination. The system performs multi-factor analysis by jointly modelling disorder-related signals and these behavioural root causes, instead of treating them in isolation. The dual-model reasoning are used to combine a BERT-based disorder screener with a classical machine-learning root-cause classifier within a single decision pipeline.

The system was fully implemented in Python (version 3.10) and executed in both local and cloud environments. Deep learning components were trained using PyTorch and Hugging Face Transformers, while classical machine-learning models were built with scikit-learn, employing TF-IDF vectorization and One-vs-Rest Logistic Regression classifiers. Supporting data operations utilized NumPy and Pandas, and model artefacts were serialized via joblib for reproducible deployment. The conversational interface was developed in Streamlit, coordinating real-time interaction between users, machine-learning models, and the OpenAI SDK, which generated natural reflective prompts aligned with cognitive-behavioural therapy (CBT) tone. All experimental runs were executed deterministically with fixed random seeds, and model outputs, metrics, and configurations were stored under a structured directory hierarchy for reproducibility.

Training and evaluation were conducted on Google Colab Pro using an NVIDIA Tesla T4 (16 GB) GPU, while local validation ran on CPU/GPU workstations supporting BERT-base and DistilBERT configurations with sequence lengths up to 512 tokens and batch sizes between 8–16. The training corpora comprised both disorder-screening and procrastination-cause datasets. For disorder detection, separate corpora were used for depression and anxiety. The depression dataset was balanced across positive and negative samples, whereas the anxiety corpus initially contained only positive examples. Neutral sentences were therefore added as synthetic negatives, and explicit anxiety terms were removed through regular-expression filtering to prevent label leakage. Data were divided into fixed train/validation/test splits (typically 80/10/10 or 70/15/15) with consistent random seeds. A custom, partly synthetic root-cause dataset was prepared to capture nonclinical triggers of procrastination—such as perfectionism, fear of failure, lack of interest, environmental distraction, dopamine addiction, and feeling overwhelmed—encoded as one-hot multi-label targets for training and evaluation. Text pre-processing was intentionally minimal to preserve emotional nuance. Whitespace and redundant spacing were removed, but punctuation and expressive vocabulary were

retained to maintain tone and affect. Conversation transcripts were prefixed with speaker identifiers [USR] and [SYS] to distinguish turns during model training and inference. Tokenization used the BERT/DistilBERT tokenizer with dynamic padding and truncation to 256–512 tokens per sequence. For long conversations, overlapping sliding windows (stride ≈ 64 tokens) preserved contextual continuity during both training and inference.

The primary deep-learning module was a Two-Head BERT model for disorder screening. It employed a shared encoder whose [CLS] representation fed two independent output heads predicting depression-like and anxiety-like probabilities. Each head consisted of a linear transformation followed by a sigmoid activation, trained using binary cross-entropy with logits. To handle class imbalance, pos weight adjustments and Focal Loss ($\gamma = 2.0$, $\alpha = 0.75$) were explored to penalize easy negatives. Two operational modes were implemented: a quick mode that concatenated recent dialogue turns into a single text block, and a conversation-aware mode retrained on full multi turn transcripts tagged with [USR]/[SYS]. Output thresholds were tuned between 0.1–0.9 to maximize the F1 score on validation data. Optimization used AdamW with a learning rate of 2×10^{-5}, linear warm-up, gradient clipping (1.0), and mixed precision where available. Models typically converged within 2–4 epochs, alternating between depression and anxiety batches to maintain balanced updates.

The secondary analytical module was a Root-Cause Classifier aimed at interpreting motivational and contextual drivers of procrastination. Text inputs were vectorised with TF-IDF (unigrams + bigrams) and passed through One-vs-Rest Logistic Regression models trained with class weights inversely proportional to label frequency. Each class threshold was individually calibrated to maximize F1 score, and all artefacts—including vectorizers, models, and threshold files—persisted for inference. The end-to-end workflow was orchestrated through a Streamlit interface managing user state, chat history, and safety mechanisms. The main interface displayed chat dialogues, while a sidebar offered screening tools for disorder and root-cause analysis. Incoming text was routed through TwoHeadInfer.predict() for disorder evaluation and through rc_infer.predict_root_causes() for motivational interpretation. All communication occurred within the user session, ensuring no personal data were stored. If a conversation indicated self-harm or violence, the system paused interaction and displayed crisis-support resources. Model inference combined multiple optimizations for re liability and interpretability. For conversation-aware BERT inference, dialogues were truncated to ∼1800 characters and segmented using a stride of 64 tokens (max len = 512). Window logits were temperature-scaled ($T \approx 1.5$) to avoid over confidence and aggregated through 80th-percentile pooling, balancing stability and sensitivity. For root-cause inference, early messages were vectorised with the saved TF-IDF model and evaluated through the logistic regressor; probabilities above calibrated thresholds were flagged as active causes.

Evaluation followed standard machine-learning metrics. The models were assessed on ROC-AUC, PR-AUC, and per-head Precision, Recall, and F1 scores at tuned thresholds. Ablation studies compared last-message versus conversation-level analysis, various pooling strategies (max, mean, percentile), and temperature-scaling settings ($T = 1.0–2.0$). Manual inspection of false predictions identified common error patterns such as misinterpreting enthusiasm as anxiety or misclassifying system messages as user input. The overall system design prioritized user safety, fairness, and transparency. Bias mitigation combined balanced data sampling, Focal Loss, adaptive thresholds, percentile pooling, and temperature scaling. Ethical safeguards ensured that the tool provided reflective support rather than medical diagnosis, never storing personal identifiers or session logs. Mentions of self-harm automatically triggered predefined safety messaging. Model behaviour was continuously monitored, and prediction trends were periodically recalibrated to maintain fairness across user segments. Reproducibility was ensured through deterministic seeding, TABLE I version-controlled configurations, and standardized directory structures. Each training run produced a complete artifact package containing model weights, tokenizer files, thresholds, and logs. Models were dynamically loaded at runtime through load_screening_model() for the BERT component and rc_infer.py for root-cause classification, allowing updates without code modification. This modular design ensured that all system components—from data pre-processing to inference—remained transparent, explainable, and ready for real world deployment. These fifteen studies show a pattern. Early studies used simple models to diagnose problems. They also used basic chatbots. Later studies combined these features into smarter, adaptable systems. Most projects focus on either predicting issues or providing therapy. Few projects do both. Our study, Procrastination Root Cause Identifier (MDoc), combines these approaches. It uses double AI models, offers Cognitive Behavioural Therapy (CBT) guidance, and adds game-like

motivation. This creates a system where users get personalized help before problems develop. The work moves from just detecting issues to actively helping users improve their mental health.

4. Results and Analysis

After building and refining the Procrastination Coach, the evaluation has been conducted how well it understood emotional tone and motivational patterns in real-world conversations. The section shares what we learned: the performance of the conversation-aware two-head BERT model in screening for depression and anxiety, insights from the multi-label root-cause classifier, and how interpretability as well error analysis shaped the understanding of the system's behaviour. All metrics refer to the held-out test set unless specified otherwise. These patterns are consistent with earlier psychological studies showing that procrastination arises from both emotional distress and behavioural root causes, rather than any single trigger. The model's stronger performance on conversation-level inputs also supports the view that understanding these patterns requires multi-factor analysis instead of isolated cues. Overall, the results reinforce the value of using dual-model reasoning to examine mood indicators and motivational drivers together.

4.1 Ablation Study

Purpose: The goal is to understand the true contribution of each modelling choice—conversation context, pooling strategy, loss function, and threshold calibration. To keep comparisons fair, all experiments used identical data splits, seeds, and optimizers; only one variable changed at a time.
Baseline: We began with a simple model that looked only at the last message of each conversation. It used a plain BCEWithLogitsLoss with no class weighting or focal adjustments, and a fixed threshold of 0.5. This version worked reasonably well for obvious emotional statements but struggled whenever the emotional signal was subtle or appeared earlier in the dialogue. It often missed quiet forms of anxiety and low-energy descriptions of depression.

Step-by-Step Improvements:
Takeaway: The greatest leap came from adding conversation context and robust pooling. Focal loss and adaptive thresholds further refined behaviour—reducing over-sensitivity without blunting empathy as depicted in Table.1.

Table 1. Ablation Steps and Effect

Step	Change Introduced	Observed Effect
I	Added context by joining turns and tokenized in 512-token windows	Recall improved for depression/anxiety; more FP in short/noisy snippets.
II	Robust ignore pooling: small windows, use 80th percentile pooling.	Precision improved for anxiety; one-word spikes vanished.
III	Temperature scaling (T =1.5).	Outputs smoother; better calibration.
IV	Class weighting (pos_weight).	Recall improved; F1 scores steadier.
V	Focal Loss ($\gamma = 2.0$, $\alpha = 0.75$).	FPs, esp. for anxiety, dropped; handled borderline phrases.
VI	Per-head threshold calibration.	Most balanced; each head optimal thresholds.

Final Calibrated Test-Set Scores: Why different thresholds? Depression often reveals itself through muted language, spread thinly across messages; a lower threshold ($t* = 0.24$) helps catch softer cues. Anxiety uses sharper language and overlaps with ordinary excitement; a higher threshold ($t* = 0.80$) prevents misreading positive energy as anxiety as shown in Table.2 and Table.3.

Table 2. Final Model Performance of Disorder Causes

Model Head	Accuracy	Precision	Recall	F1	t*
Depression	0.9817	0.9866	0.9761	0.9813	0.24
Anxiety	0.9906	0.9893	0.9919	0.9906	0.80

Table 3. Final Model Performance of Non-Disorder Causes

Model Head	Accuracy	Precision	Recall	F1	t*
Perfectionism	0.925	1.000	0.833	0.909	0.90
Fear of Failure	0.889	0.800	1.000	0.889	0.20
Lack of Interest	0.857	0.750	1.000	0.857	0.40
Environmental Distraction	0.800	0.769	0.833	0.800	0.05
Dopamine Addiction	0.727	0.571	1.000	0.727	0.20

4.2 Comparative Model Performance and Interpretability

Quantitative Comparison: The curves shown in Fig.2, Fig.3, Fig.4 and Fig.5 summarize discrimination quality for both the disorder screening heads and the multi-label root-cause classifier.

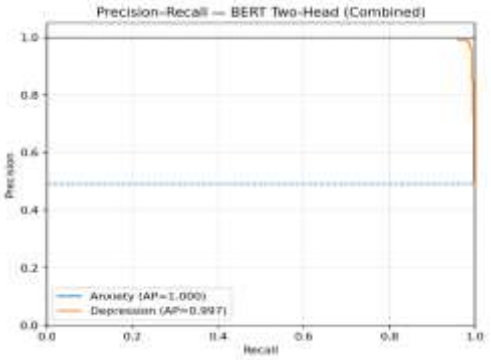

Fig. 2. Precision- Recall - BERT Two-Head Model

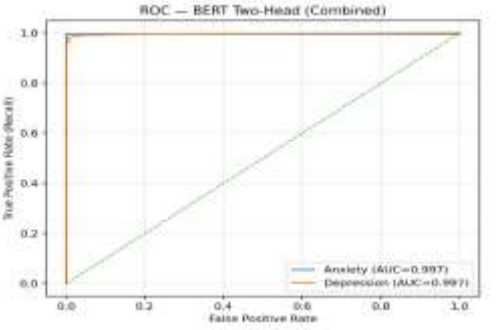

Fig. 3. ROC — BERT Two-Head Model

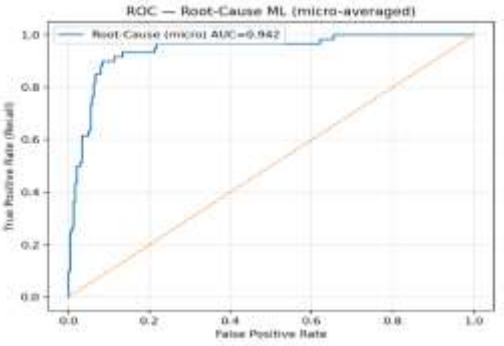

Fig. 4. ROC - Root-Cause Classifier

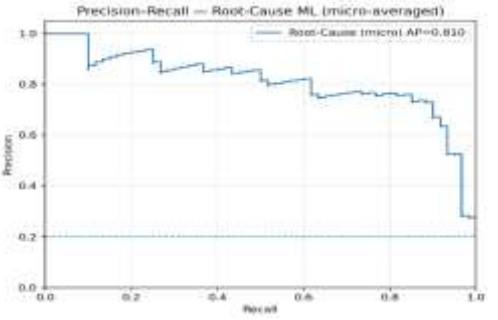

Fig. 5. Precision–Recall -Root-Cause Classifier (micro-averaged)

Interpretability and Error Analysis:

Gradient-based attributions highlighted where the BERT model focuses:
• Depression: “nothing feels worth it,” “no energy,” “lost interest,” “everything is heavy”—often distributed across turns.
• Anxiety: “racing thoughts,” “what if I fail,” “tight chest,” “panic,” and temporal markers (e.g., “at night”).

Common failure modes:

- FP (anxiety): Highly enthusiastic lines (e.g., “I’m super excited!!!”) or assistant-only snippets.
- FN (depression): Understated single-turn remarks (e.g., “it’s fine, I guess”); conversation aggregation mitigated this.
- Overlap: Mixed affect sometimes activates both heads; acceptable for screening, with both scores shown in the UI.

What truly made a difference:

- Conversation context + percentile pooling delivered the biggest gains.
- Focal loss and head-specific thresholds balanced sensitivity and precision (empathy vs. caution).
- The root-cause module added interpretability and action ability beyond binary screening.

Practical and UX implications

- The tool supports reflection and early screening, not clinical diagnosis.
- Showing calibrated probabilities and thresholds helps users understand uncertainty.
- Root-cause probabilities align interventions with likely drivers (e.g., perfectionism, dopamine-seeking).

Limitations and future work

- Context length: 512-token limits required windowing; longer-context models are a natural next step.
- Label noise: Public data approximate clinical labels; periodic expert validation is needed.
- Domain drift: Language evolves; scheduled re-training is important.
- Joint modelling: Unifying disorder heads and root causes in a multi-task setup could improve generalization.

5. Conclusion

The Procrastination Root Cause Identifier (PRCI) research tries to understand why people procrastinate by examining emotions and habits using artificial intelligence. It helps to determine if someone’s delay is related to mental health issues like anxiety or depression, or more general factors such as fear of failure, perfectionism, or low motivation. By combining deep learning and machine learning through dual-model reasoning, PRCI provides a system that is both effective and empathetic. The BERT model finds emotional cues, while TF-IDF and Logistic Regression models detect each behavioural root cause, enabling clearer multi-factor analysis of user conversations. The interface, built in Streamlit and connected to a Large Language Model, allows users to engage in natural dialogue and access clear results. Experimental testing shows that PRCI can reliably distinguish procrastination linked to disorders from ordinary motivational patterns, validating the implementation choices and model design. Users receive real-time information about their emotional and behavioural drivers, which promotes self-awareness and reflection. The system prioritizes privacy, does not retain personal information, and offers supportive suggestions when users express stress or distress. Overall, PRCI highlights how AI can be used responsibly to understand human behaviour. In the future, the system could be tested with real users to study comfort, accuracy, and long-term impact, demonstrating how AI-supported feedback might help people improve their habits and emotional well-being in a safe, private, and supportive manner. Future work could introduce a gamified habit-building module that recommends personalised tasks based on the user’s identified procrastination causes, supports progress tracking, and helps reduce addictive patterns. A linked therapist dashboard could also deliver weekly summaries and reported challenges, enabling timely professional support when needed.

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